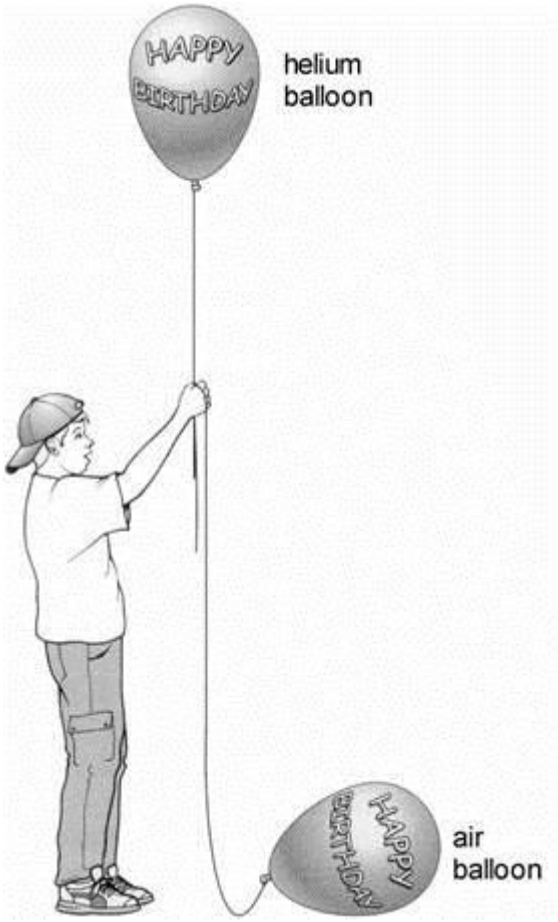
Q1.

Chris has two rubber party balloons. One is filled with air and the other is filled with helium. Both balloons contain the same volume of gas.



(a) (i) Explain why the helium balloon rises.

.....

.....

1 mark

(ii) Explain why the air balloon drops to the ground.

.....

.....

1 mark

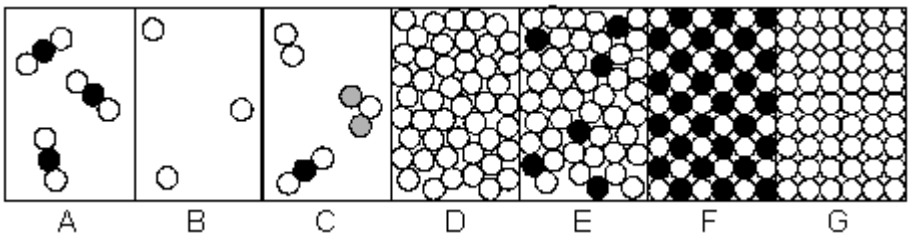
(b) The chemical symbol for helium is He.
Explain why air does **not** have a chemical symbol or formula.

.....

.....

1 mark

(c) The diagram below shows seven arrangements of particles.



The two party balloons are coated with a thin layer of aluminium.
Give the letter of the diagram which best represents the particles in:

(i) the helium gas; .....

1 mark

(ii) the air; .....

1 mark

(iii) the thin layer of aluminium. .....

1 mark

(d) Over several days, the balloons shrink because the particles of gas diffuse through the balloon and escape. The helium balloon shrinks more quickly than the air-filled balloon. Answer the following questions in terms of particles.

(i) Why does helium escape more quickly than air from a balloon?

.....

.....

1 mark

(ii) A rubber balloon coated with aluminium takes longer to shrink than a rubber balloon **without** an aluminium coating.
Suggest a reason why gas particles diffuse more slowly through aluminium than through rubber.

.....

.....

1 mark
Maximum 8 marks

Q1.

(a) (i) helium is less dense than air

accept 'helium is lighter than air'

accept 'the upthrust on the helium balloon is greater than the weight of the balloon'

1 (L7)

(ii) any **one** from

- the air in the balloon is denser than the air in the room

  accept 'the air in the balloon is compressed'

- the rubber has weight

  accept 'rubber is heavier than air'

  accept 'the rubber is denser than air'

  accept 'the upthrust on the air balloon is less than the weight of the balloon'

1 (L7)

(b) any **one** from

- air is a mixture

  accept 'air contains different gases'

- air is not a single element **or** compound

  'air is not a single substance' is not sufficient

1 (L7)

(c) (i) B

1 (L7)

(ii) C

1 (L7)

(iii) G

1 (L7)

(d) (i) any **one** from

- helium particles are smaller

  accept 'molecules' or 'atoms' for particles

- helium particles move faster

1 (L7)

(ii) any **one** from

- aluminium particles **or** atoms are closer together

- rubber particles **or** molecules are further apart

  accept 'rubber particles
  ***or** molecules have bigger gaps between them'*

1 (L7)

[8]